(b) This formula is due to Sylvester, which enable one to compute the projections directly.

* The advantage of this formula is that one doesn't need to know the eigenvectors in order to find the projections.

* Since A is a Hermitian operator, suppose it's represented by a $n \times n$ matrix, and change eigenvalue a_i to λ_i or λ_j.

$$A = \sum_j \lambda_j |\lambda_j\rangle\langle\lambda_j| \qquad i, j \in \{1, 2, \dots n\}$$

$$= \lambda_i |\lambda_i\rangle\langle\lambda_i| + \sum_{j\,(i\neq j)} \lambda_j |\lambda_j\rangle\langle\lambda_j|$$

$$= \lambda_i \left(I_{n\times n} - \sum_{j\,(i\neq j)} \lambda_j |\lambda_j\rangle\langle\lambda_j|\right) + \sum_{j\,(i\neq j)} \lambda_j |\lambda_j\rangle\langle\lambda_j|$$

$$= \lambda_i I + \sum_{j\,(i\neq j)} (\lambda_j - \lambda_i) |\lambda_j\rangle\langle\lambda_j|$$

$$(A - \lambda_i I) = \sum_{j\,(i\neq j)} (\lambda_j - \lambda_i) |\lambda_j\rangle\langle\lambda_j|$$

By the Mathematical induction we can show that

$$\prod_{i=1}^{n} (A - \lambda_i I) = (\lambda_n - \lambda_1)(\lambda_n - \lambda_2) \cdots (\lambda_n - \lambda_{n-1}) |\lambda_n\rangle\langle\lambda_n|$$

$$\Rightarrow (A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_{\ell-1} I)(A - \lambda_{\ell+1} I) \cdots (A - \lambda_n I)$$

$$= (\lambda_\ell - \lambda_1)(\lambda_\ell - \lambda_2) \cdots (\lambda_\ell - \lambda_{\ell-1})(\lambda_\ell - \lambda_{\ell+1}) \cdots (\lambda_\ell - \lambda_n) |\lambda_\ell\rangle\langle\lambda_\ell|$$

$$\vdots \quad j = n,\ i \neq 1, 2, \dots, n-1$$

$$\Rightarrow \prod_{\substack{i=1 \\ i\neq l}}^{n} (A-\lambda_i I) = \left[\prod_{\substack{i=1 \\ i\neq l}}^{n} (\lambda_l-\lambda_i)\right] |\lambda_l\rangle\langle\lambda_l|$$

$$\Rightarrow \boxed{|\lambda_l\rangle\langle\lambda_l| = \frac{\prod_{i\neq l}^{n} (A-\lambda_i I)}{\prod_{i\neq l}^{n} (\lambda_l-\lambda_i)}} = \frac{\prod_{i\neq l}^{n} (A-a_i I)}{\prod_{i\neq l}^{n} (a_l-a_i)}$$

Now, for a $n\times n$ matrix A with distinct eigenvalues λ_j and a function f, so that the spectral decomposition law may be expressed as

$$P(A) = \sum_{j=1}^{n} f(a_i) \frac{\prod_{i\neq j} (A-a_i I)}{\prod_{i\neq j} (a_i-a_j)}$$

if we let

$$P(A) = \sum_i P(a_i) |\lambda_i\rangle\langle\lambda_i|$$

where $A|a_i\rangle = a_i|a_i\rangle$, $i=1,2,\ldots n$

then

$$P(A) = \sum_l P(a_l) |\lambda_l\rangle\langle\lambda_l| \quad \text{——①}$$

$$= \sum_l P(a_l) \frac{\prod_{i\neq l}^{n} (A-a_i I)}{\prod_{i\neq l}^{n} (a_l-a_i)} \quad \text{——②}$$

and compare with ① and ② we conclude that we don't need to know eigenvectors when eigenvalues are distinct, but still can find the projections!